Table 1. With ACTOR, the Compliance Rate on false refusal datasets increases across all the models while keeping the model safe.

Models	Method	XS Test-Safe C.R	SCOPE C.R	OR-Bench Hard C.R	PHTest C.R	OKTest C.R	Avg. C.R	AdvBench S.S	Avg. TS ↑
Llama-2-7b-chat	Default	80	52.61	29.45	69.60	76.0	61.53	99.62	80.58
	ACTOR	95.33	91.57	76.28	96.86	93.67	90.74	99.03	**94.89**
Gemma-7b-it	Default	72.00	58.18	65.71	88.92	74.00	71.76	94.00	82.88
	ACTOR	79.33	62.73	73.83	91.15	78.00	77.01	92.5	**84.75**
Llama-13b-chat	Default	79.33	44.69	23.54	76.13	78.67	60.47	100	80.23
	ACTOR	87.33	81.28	65.30	95.93	90.67	84.10	99.62	**91.86**

Table 2. Performance Comparison of ACTOR with baselines. Best results highlighted in **bold**.

Models	Method	XS Test-Safe C.R	SCOPE C.R	OR-Bench Hard C.R	PHTest C.R	OKTest C.R	Avg. C.R	AdvBench S.S	Avg. TS ↑
Llama-2-7b-chat-hf	SFT	87.33	86.68	62.18	79.56	90.67	81.28	99.03	90.16
	Safe-Decoding	29.12	15.32	7.45	26.03	45.24	24.63	100.00	62.32
	DRO	58	21.25	14.11	62.22	76.00	46.32	100.00	73.16
	Self CD	90.67	80.94	61.94	87.41	92.00	82.59	95.77	89.18
	SCANs	95.33	76.72	40.52	90.44	99.00	80.40	99.23	89.82
	Surgical	90.67	89.38	69.16	93.42	89.33	86.39	99.42	92.90
	Ours	95.33	91.57	76.28	96.86	93.67	**90.74**	99.03	**94.88**
Gemma-7b-it	SFT	72.00	58.52	67.35	89.43	75.33	72.53	94.2	83.37
	Safe-Decoding	32.12	19.43	8.32	38.21	40.34	27.68	98.32	63.00
	DRO	52.04	44.92	58.39	75.01	71.28	60.33	97.78	79.06
	Self CD	78.00	64.75	74.20	88.08	73.00	75.61	87.12	81.36
	SCANs	56.66	56.15	70.87	80.12	53.66	63.49	93.65	78.57
	Surgical	76.67	61.20	74.20	89.72	76.33	75.62	90.96	83.29
	Ours	79.33	62.73	73.83	91.15	78.00	**77.01**	92.5	**84.75**

Metrics. Over-refusal is measured using the **Compliance Rate (C.R)**, defined as the ratio of compliant responses to the total number of generated responses. Similarly, the model's safety is evaluated with the **Safety Score (S.S)**, calculated as the ratio of denials to the total number of responses. To evaluate the trade-off between safety and over-refusal, we introduce the **Tradeoff Score (T.S)**, calculated as the average of the compliance rate and safety score. Response evaluation for compliance and safety was conducted using GPT-4o (Appendix E). For MT-Bench, responses are scored on a scale of 1 to 10. In the multiple-choice setting of MMLU, we evaluate accuracy across four categories and report the overall average score.

Models and Hyperparameters. The goal of our study is to develop a lightweight method to correct over-refusal behaviors in safety-aligned models. We choose three open-source aligned models- Llama-2-7b-chat, Llama2-13b-chat (Touvron et al., 2023) and Gemma-7b-it (Team et al., 2024). In each experiment, we train the models for 3 epochs using the AdamW optimizer (Loshchilov, 2017). For Supervised Fine Tuning (SFT), a learning rate of 1e-3 was employed while experiments using ACTOR used a learning rate of 1e-5.

4.2. Experimental Results

ACTOR effectively mitigates exaggerated safety. As shown in Tables 1 and 2, ACTOR effectively achieves a balance between exaggerated safety mitigation and adequate safety and outperforms all baseline methods. Specifically, ACTOR improves the average Compliance Rate by **47.47%** for Llama-2-7b-chat, **39.07%** for Llama-2-13b-chat, and **7.31%** for Gemma-7b-it. Moreover, the Safety Scores on AdvBench demonstrate that ACTOR has almost no influence on the maintenance of adequate safety. To fine-tune with ACTOR, we tested multiple values of the hyperparameter α to identify the optimal configuration, with the best results shown in Table 1. An ablation study analyzing the impact of α on performance is provided in Appendix C.2.

Table 3. Train-Time Comparison of fine-tuning LLama-2-7b-chat with SFT and ACTOR- 3 epochs on H100

Method	Train Time
SFT	15 mins
ACTOR	4 mins

ACTOR is both Compute and Data-Efficient. Unlike traditional training-based methods that require retraining the entire model, ACTOR achieves superior performance by fine-tuning just a single layer. This approach not only reduces computational overhead but also delivers stronger and more precise learning signals for controlling the refusal mechanism. Table 3 presents a comparison of the training times between SFT and ACTOR. An equally compelling aspect of ACTOR is its data efficiency. Unlike response-driven training algorithms, ACTOR operates in a response-free setting, relying solely on queries. As shown in Figure 4 (Left), ACTOR consistently outperforms standard SFT across varying data budgets, even when the dataset includes as few as 25 pseudo-harmful queries.

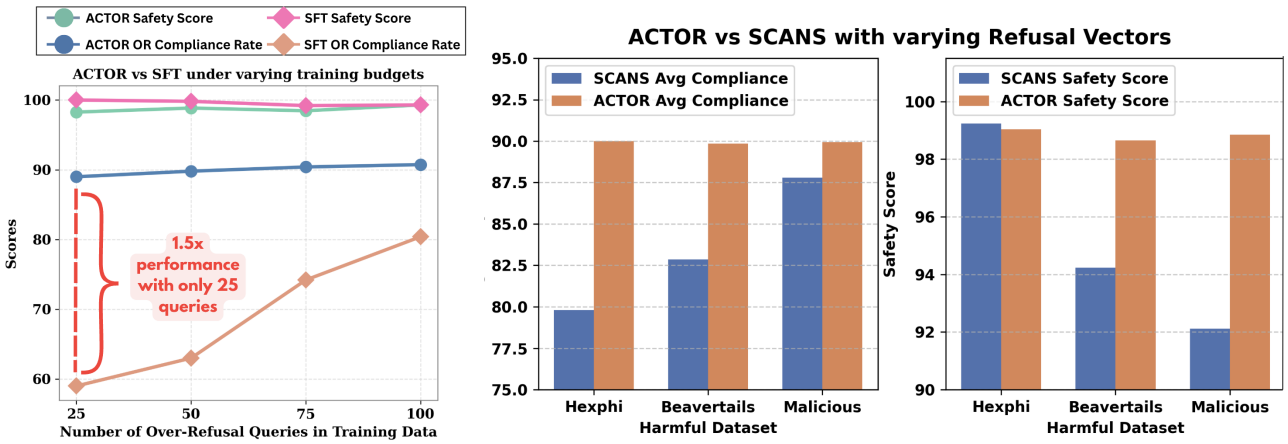

Figure 4. **(Left) Comparison of ACTOR and SFT Across Varying Data Budgets**: The number of over-refusal queries in the training data is varied, while the number of safe and harmful queries remains constant. **(Right) Robustness of ACTOR:** This figure compares the performances of SCANS and ACTOR using Refusal Directions computed from different harmful distributions, while the benign dataset used is TruthfulQA in all cases. The consistent performance of ACTOR highlights its robustness under distributional shifts.

Table 4. The impact of ACTOR on general model capability.

Model	MT Bench ↑			Perplexity ↓	MMLU ↑				
	1 Turn	2 Turn	Avg.	WikiText2	STEM	Humanities	Social Sciences	Others	Avg.
Llama-2-7b-chat	6.81	5.67	6.23	11.59	36.09	43.27	53.03	54.84	46.81
+ **ACTOR**	6.65	5.65	6.16	11.57	36.44	43.40	53.26	55.10	47.05
Gemma-7b-it	7.23	5.62	6.43	38.38	42.65	44.88	58.07	57.93	50.88
+ **ACTOR**	7.32	5.33	6.33	37.29	42.97	45.12	58.66	58.41	51.29
Llama-13b-chat	7.15	6.20	6.67	10.03	42.68	49.73	61.42	60.63	53.61
+ **ACTOR**	7.16	6.18	6.67	10.00	42.84	49.47	61.39	60.44	53.54

ACTOR is Distributionally Robust. Existing single-vector methods such as SCANS (Cao et al., 2024) and Surgical (Wang et al., 2024) are highly sensitive to the data used when computing their refusal directions. For instance, keeping the benign set fixed to TruthfulQA (Lin et al., 2021) but swapping the harmful set among HexPhi (Qi et al., 2023), BeaverTails (Ji et al., 2024), and MaliciousQA (Huang et al., 2023) causes SCANS to fluctuate markedly, producing large variance in both Compliance Rate and Safety Score as shown in Figure 4 (Right). Surgical shows a similar fragility: moving from an ORBench-based direction to one derived from OKTest drops its average over-refusal Compliance Rate from 86.39% to 63.88%.

ACTOR avoids this brittleness by optimizing the model parameters directly rather than committing to a single static vector. During fine-tuning, ACTOR iteratively updates both the parameters and the current refusal direction. Instead of applying a fixed global shift, its loss encourages a *query-dependent* adjustment—proportional to each example's projection onto the refusal direction—yielding only the "just-enough" intervention required. This dynamic scheme allows ACTOR to leverage the full capacity of the model's activation space, maintaining stable performance even as the underlying data distribution changes.

Consequently, ACTOR is more than a straightforward fine-tuned version of prior single-vector approaches: it is explicitly designed to overcome their sensitivity to dataset choice and delivers a more adaptive and reliable alignment strategy.

Effects on Model Capability: Table 4 presents the evaluation of general model capabilities before and after fine-tuning LLMs with ACTOR. The MTBench score remains relatively stable, with a drop of no more than 1.5%. Additionally, models fine-tuned using ACTOR demonstrate consistently lower perplexity across all cases, while the percentage change in accuracy for the MMLU task stays within 1%. These results demonstrate that ACTOR minimally impacts general model capabilities while maintaining its effectiveness.

5. Limitations and Future Work

This study is constrained by several factors that shape both its conclusions and avenues for future work. Because we evaluate over-refusal with multiple public benchmarks—each designed by different stakeholders who hold their own views on what constitutes "safe" content—label disagreements inevitably arise: a query tagged as pseudo-harmful in ORBench-Hard-1K, such as asking for illicit trading techniques, might be deemed fully disallowed by others, and likewise, chemical-safety discussions can be benign for specialists yet concerning for lay audiences. Such inconsistencies mean our results reflect a compromise among the normative preferences embedded in these fixed datasets rather than a universal notion of safety. In addition, ACTOR operates in a white-box setting: it relies on direct access to model activations and on an existing safety-alignment mechanism to compute a refusal direction, resources that third-party users of strict black-box APIs lack. While this requirement limits immediate external adoption, model providers themselves do control these internals and could implement ACTOR, and the representational visibility it affords offers safety insights that prompt-only methods cannot capture. Looking ahead, an important direction is to close the loop by jointly curating data and fine-tuning the model—rather than optimizing against a fixed collection of benchmarks—and to extend evaluation to multi-turn scenarios, where over-refusal and genuine safety concerns intersect in realistic dialogue yet remain largely unmeasured due to the absence of dedicated benchmarks.

6. Conclusion

In this work, we introduced **ACTOR**, a novel activation-based training framework designed to mitigate over-refusals in LLMs while preserving safety and utility. At the core of ACTOR is our proposed *Projection-Calibrated Refusal Direction Loss*, which precisely adjusts model activations to ensure necessary shifts in the refusal mechanism. Unlike traditional response-based methods, ACTOR directly optimizes internal activation patterns, enabling stronger control over safety alignment. Our approach is inherently **data-efficient**, requiring no additional response supervision, and **compute-efficient**, fine-tuning only a single layer of the model. By effectively balancing over-refusal reduction and safety preservation across multiple benchmarks, ACTOR highlights the potential of activation-based fine-tuning as a powerful alternative to traditional methods.

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Impact Statement

This paper presents a novel fine-tuning approach for mitigating over-refusals in Large Language Models (LLMs) while maintaining strong safety alignment. Our method specifically addresses the issue of over-refusals, where models learn to overgeneralize refusals to benign pseudo-harmful queries. By leveraging activation-based training, we systematically reduce unnecessary denials while ensuring that genuinely harmful queries still elicit appropriate refusals.

The broader societal impact of this work is twofold. On one hand, it enhances the usability of LLMs by reducing excessive safety guardrails that can hinder access to legitimate information. Secondly, it prevents potential misuse by ensuring that harmful prompts continue to be reliably refused. However, as with any safety fine-tuning method, further evaluations in real-world settings are necessary to assess long-term robustness against adversarial exploitation.

By advancing safe and aligned AI, this work contributes to the responsible deployment of generative models in applications that demand nuanced, context-aware moderation.

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